Chapter 2

E-business strategy

Learning Objectives

In this chapter, you will learn about:

1. Introduction to the strategy of E-business
2. Web server and E-mail technologies
3. Revenue models of E-business
4. Online shopping
5. Virtual communities
6. E-marketing strategy and planning

3. Revenue models of E-business

What is a business model?

- A business model is a set of planned activities (sometimes referred to as business processes) designed to result in a profit in a marketplace.
- A business model is not always the same as a business strategy although in some cases they are very close as the business model explicitly takes into account the competitive environment.
- A business model is at the center of the business plan.
- A business plan is a document that describes a firm’s business model.

Eight key elements of a business model

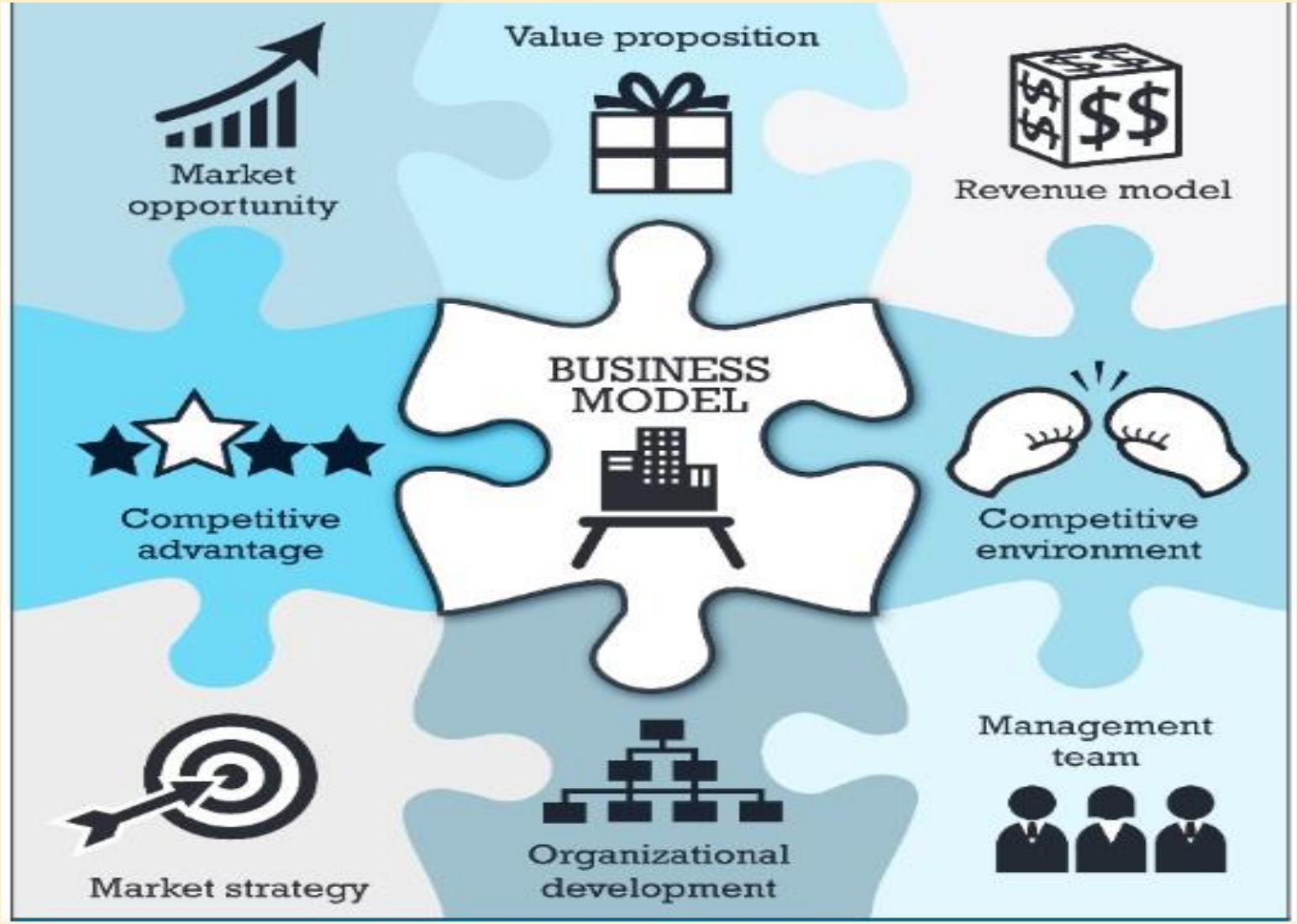


Figure 18: Eight key elements of a business model.

First element: Value Proposition

- "Why should the customer buy from you?"
- Successful e-commerce value propositions:
  - Personalization/customization
  - Reduction of product search and price discovery
  - Facilitation of transactions by managing product delivery

Second element: Revenue Model

- "How will you earn money?"
- Major types of revenue models:
  - Advertising revenue model
  - Subscription revenue model
    - Freemium strategy
  - Transaction fee revenue model
  - Sales revenue model
  - Affiliate revenue model
  - Etc.

Third element: Market Opportunity

- "What marketspace do you intend to serve and what is its size?"
  - Marketspace: Area of actual or potential commercial value in which company intends to operate
  - Realistic market opportunity: Defined by revenue potential in each market niche in which company hopes to compete
- Market opportunity typically divided into smaller niches

Fourth element: Competitive Environment

- "Who else occupies your intended marketspace?"
  - Other companies selling similar products in the same marketspace
  - Includes both direct and indirect competitors
- Influenced by:
  - Number and size of active competitors
  - Each competitor's market share
  - Competitors' profitability
  - Competitors' pricing

Fifth element: Competitive Advantage

- "What special advantages does your firm bring to the marketspace?"
  - Is your product superior to or cheaper to produce than your competitors'?
- Important concepts:
  - First-mover advantage
  - Complementary resources
  - Unfair competitive advantage
  - Leverage
  - Perfect markets

Sixth element: Market Strategy

- "How do you plan to promote your products or services to attract your target audience?"
  - Details how a company intends to enter market and attract customers
  - Best business concepts will fail if not properly marketed to potential customers

Seventh element: Organizational Development

- "What types of organizational structures within the firm are necessary to carry out the business plan?"
- Describes how firm will organize work
  - Typically, divided into functional departments
  - As company grows, hiring moves from generalists to specialists

Eighth element: Management Team

- "What kind of backgrounds should the company's leaders have?"
- A strong management team:
  - Can make the business model work
  - Can give credibility to outside investors
  - Has market-specific knowledge
  - Has experience in implementing business plans

B2C Business Models

- Electronic Retailer (E-tailer)
- Community provider (social network)
- Content provider
- Portal
- Transaction broker
- Market creator
- Service provider
- Etc.

B2C Models: E-Tailer

- Online version of traditional retailer
- Revenue model: Sales
- Variations:
  - Virtual merchant
  - Bricks-and-clicks
  - Catalog merchant
  - Manufacturer-direct
- Low barriers to entry

B2C Models: Community Provider

- Provide an online environment (social network) where people with similar interests can transact, share content, and communicate
  - Examples: Facebook, LinkedIn, Twitter, Pinterest, Tiktok, etc.
- Revenue models:
  - Typically hybrid, combining advertising, subscriptions, sales, transaction fees, and so on

B2C Models: Content Provider

- Digital content on the Web:
  - News, music, video, text, artwork
- Revenue models:
  - Use variety of models, including advertising, subscription; sales of digital goods
  - Key to success is typically owning the content
- Variations:
  - Syndication
  - Aggregators

B2C Models: Portal

- Search plus an integrated package of content and services
- Revenue models:
  - Advertising, referral fees, transaction fees, subscriptions for premium services
- Variations:
  - General (ie.: AOL, MSN)
  - Specialized (ie.: Sailnet - Sailing Community)
  - Search (ie.: Google, Bing)

B2C Models: Transaction Broker

- Process online transactions for consumers
  - Primary value proposition-saving time and money
- Revenue model:
  - Transaction fees
- Industries using this model:
  - Financial services
  - Travel services
  - Job placement services

B2C Models: Market Creator

- Create digital environment where buyers and sellers can meet and transact
  - Examples: eBay, Amazon
  - Revenue model: Transaction fees, fees to merchants for access
- On-demand service companies (sharing economy): platforms that allow people to sell services
  - Examples: Uber, Airbnb, Grab, etc.

B2C Models: Service Provider

- Online services
  - Example: Google-Google Maps, Gmail, Office 365, and so on
- Value proposition
  - Valuable, convenient, time-saving, low-cost alternatives to traditional service providers
- Revenue models:
  - Sales of services, subscription fees, advertising, sales of marketing data

B2B Business Models

- Net marketplaces
  - E-distributor
  - E-procurement
  - Exchange
  - Industry consortium
- Private industrial network

B2B Models: E-Distributor

- A version of retail and wholesale store, MRO (maintenance-repair-operation) goods, and indirect goods
- Owned by one company seeking to serve many customers
- Revenue model: Sales of goods
- Example: Grainger

B2B Models: E-Procurement

- Creates digital markets where participants transact for indirect goods
  - B2B service providers, SaaS and PaaS providers
  - Scale economies
- Revenue model:
  - Service fees, supply-chain management, fulfillment services
- Example: Amazon (AWS), etc.

B2B Models: Exchanges

- *Vertical digital marketplace* for direct inputs
- Revenue model: Transaction, commission fees
- Create powerful competition between suppliers
- Tend to force suppliers into powerful price competition; a number of exchanges have dropped dramatically
- Example: Go2Paper

B2B Models: Industry Consortia

- Industry-owned vertical digital marketplace open to select suppliers
- More successful than exchanges
  - Sponsored by powerful industry players
  - Strengthen traditional purchasing behavior
- Revenue model: Transaction, commission fees
- Example: SupplyOn

Private Industrial Networks

- The digital network is used to coordinate among firms engaged in business together
- Typically expand out of a large company's internal enterprise system
  - Key, trusted, long-term suppliers invited to network
- Example: Walmart's network for suppliers

How E-commerce Changes Business

- E-commerce changes industry structure by changing:
  - Rivalry among existing competitors
  - Barriers to entry
  - Threat of new substitute products
  - Strength of suppliers
  - Bargaining power of buyers
- Industry structural analysis

Industry Value Chains

- Set of activities performed by suppliers, manufacturers, transporters, distributors, and retailers that transform raw inputs into final products and services
- Internet reduces cost of information and other transactional costs
- Leads to greater operational efficiencies, lowering cost, prices, adding value for customers

Example of E-commerce and Industry Value Chains

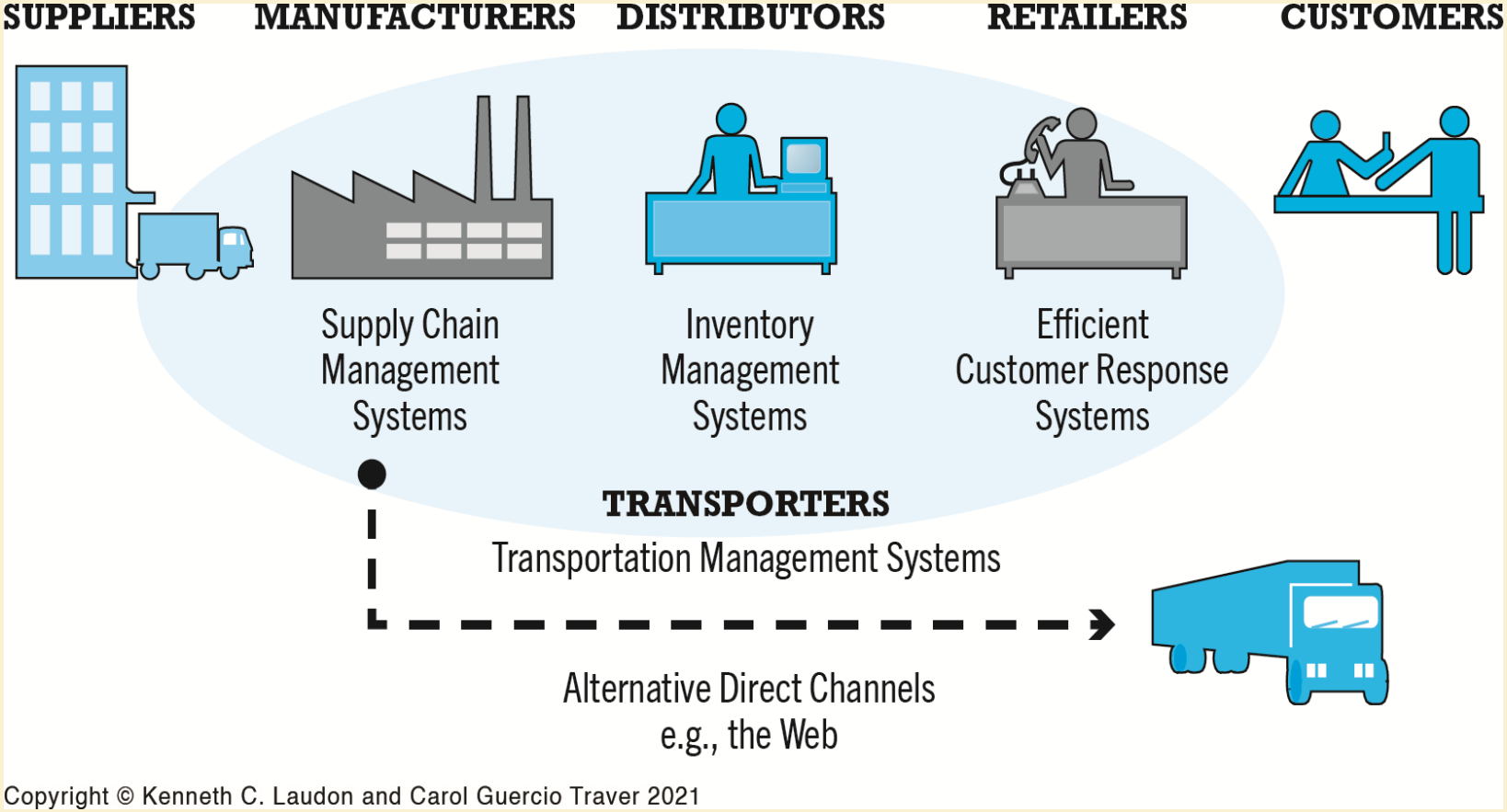


Figure 19: Example of E-commerce and Industry Value Chains.

Firm Value Chains

- Activities that a firm engages in to create final products from raw inputs
- Each step adds value
- Effect of Internet:
  - Increases operational efficiency
  - Enables product differentiation
  - Enables precise coordination of steps in chain

Example of E-commerce and Firm Value Chains

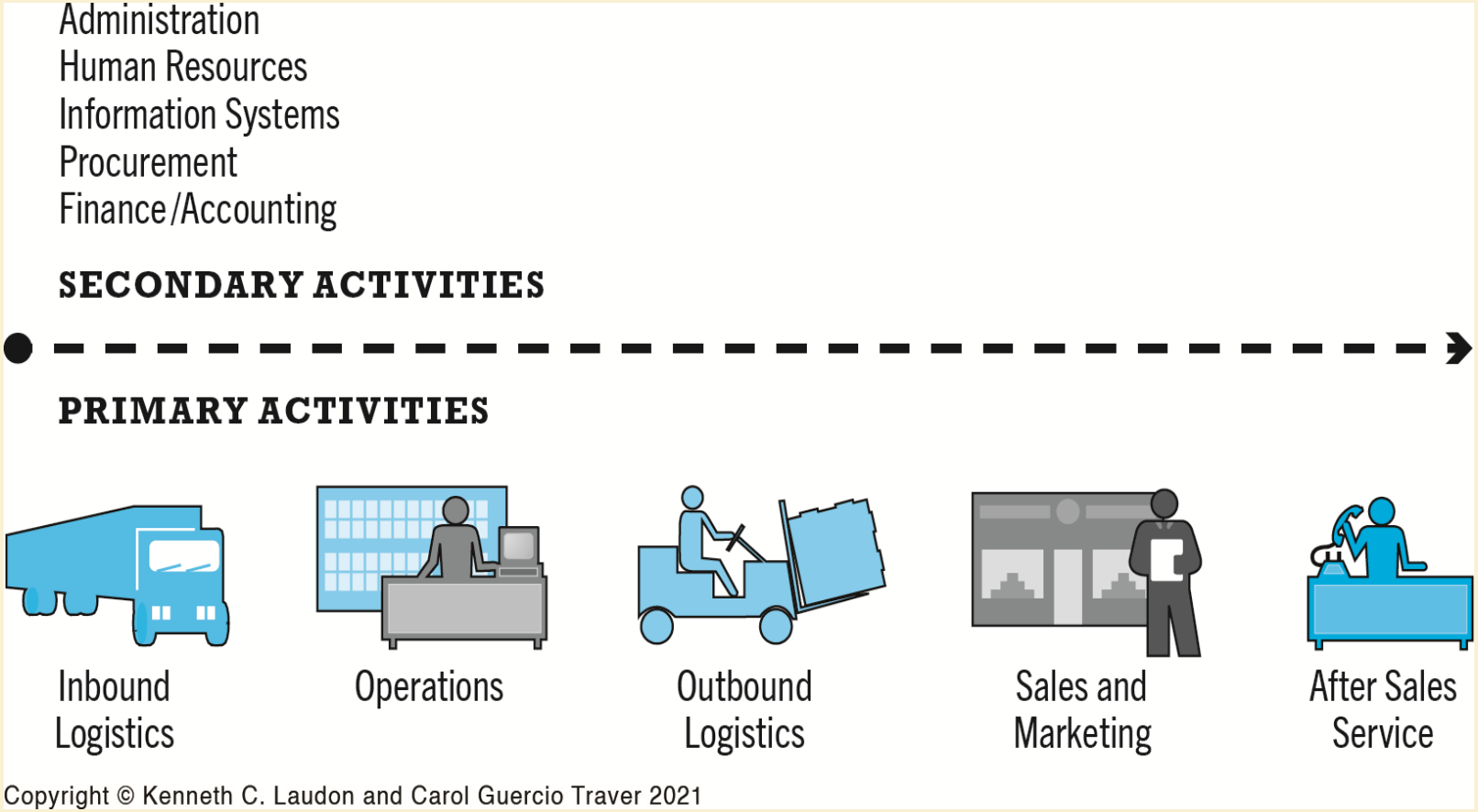


Figure 20: Example of E-commerce and Industry Value Chains.

Firm Value Webs

- Networked business ecosystem
- Uses Internet technology to coordinate the value chains of business partners
- Coordinates a firm's suppliers with its own production needs using an Internet-based supply chain management system

Internet-Enabled Value Web

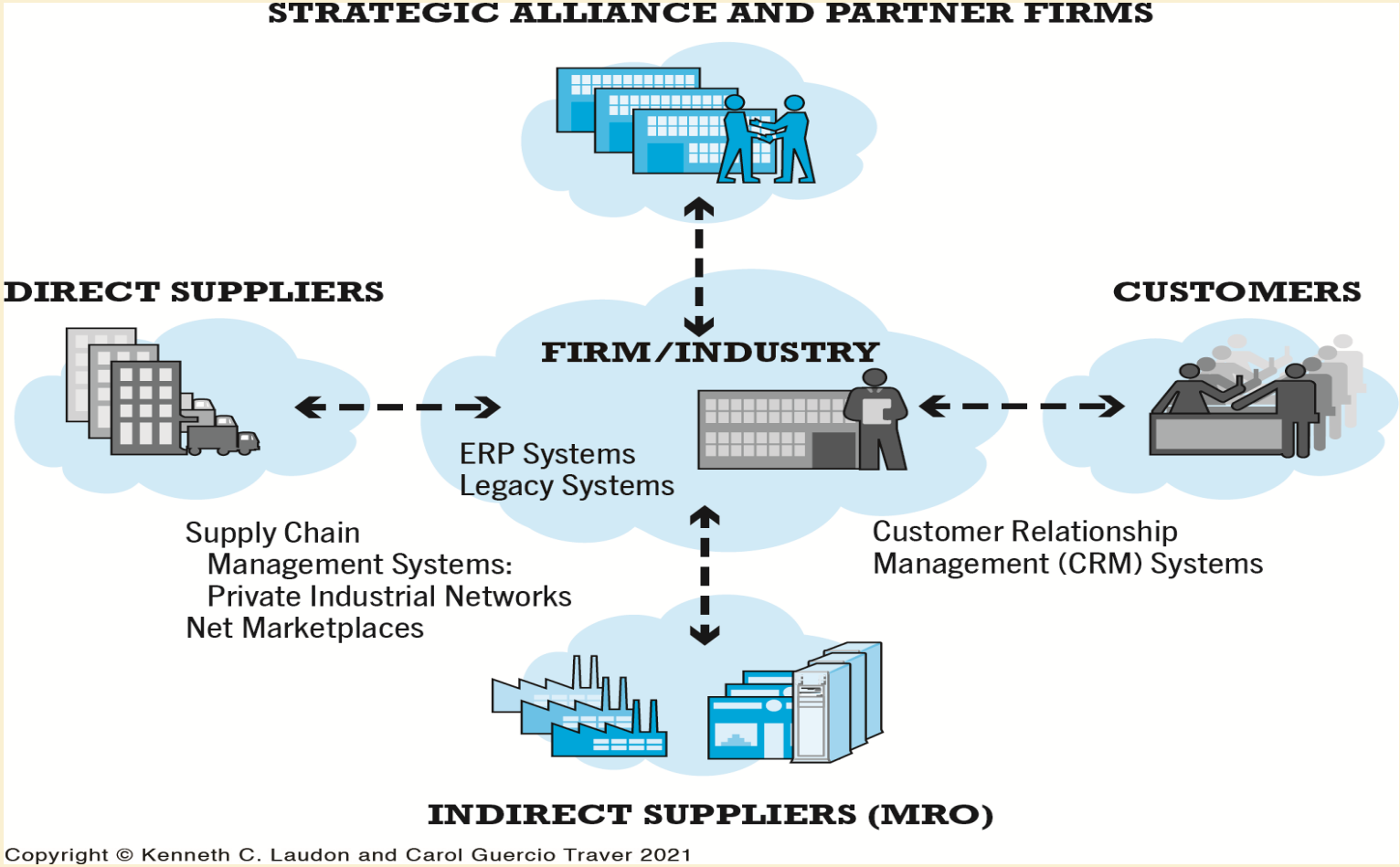


Figure 21: Internet-Enabled Value Web.

Business Strategy

- Plan for achieving superior long-term returns on capital invested: that is, profit
- Five generic strategies
  - Product/service differentiation
  - Cost competition
  - Scope
  - Focus/market niche
  - Customer intimacy/satisfaction.

E-commerce Technology and Business Model Disruption

- Disruptive technologies
- Digital disruption
- Sustaining technology
- Stages
  - Disruptors introduce new products of lower quality
  - Disruptors improve products
  - New products become superior to existing products
  - Incumbent companies lose market share

In conclusion

- What is a business model?
- Eight key elements of a business model
- B2C Business Models
- B2B Business Models
- How E-commerce Changes Business
- Example of E-commerce and Industry Value Chains
- Example of E-commerce and Firm Value Chains
- Internet-Enabled Value Web
- Firm Value Webs
- Business Strategy



4. Online Shopping

What is Online Shopping?

- Online shopping is a form of electronic commerce which allows consumers to directly buy goods or services from a seller over the Internet using a web browser or a mobile app.
- Consumers find a product of interest by searching among alternative vendors using a shopping search engine, which displays the same product's availability and pricing at different e-retailers.
- Customers can shop online using a range of different computers and devices such as desktop computers, laptops, tablet and smartphones.

Source: https://en.wikipedia.org/wiki/Online_shopping

Applied Technologies in Online Shopping

- *Chatbots and Intelligent virtual assistants (IVAs)*
  - Chatbots and IVAs are effective tools for engaging with customers and reducing engagement costs.
  - Sales chatbots and IVAs can carry out the same functions on websites, mobile applications, and messaging services like WhatsApp.

Source: https://research.aimultiple.com/ecommerce-technology/

Applied Technologies in Online Shopping (Cont.)

- *Recommendation Systems*
  - Approximately 30% of consumers, according to a PwC survey, are willing to pay more for things that are suggested to them.
  - Users are continually overloaded with suggestions. However, they are likely to appreciate personalized recommendations.
  - The overall benefit is higher rates of engagement and boosted sales.

Applied Technologies in Online Shopping (Cont.)

- *AI-driven pricing tools*

  - With the use of pricing tools, you can instantly evaluate your prices by comparing them to those of your rivals, your stock levels, delivery policies.

  - In highly competitive marketplaces, immediately changing pricing to match those employed by rivals can also significantly impact conversion rates.

Applied Technologies in Online Shopping (Cont.)

- *Web scrapping*
  - Web scraping is the process of extracting data from websites using a scraping bot or web scraping API.
  - The technique is advantageous for E-commerce businesses because these companies produce:
    - Customer data
    - Product data
    - Financial data

Applied Technologies in Online Shopping (Cont.)

- *Web scrapping*
  - Successful web scraping initiatives are beneficial for companies because they enable:
    - Price comparison with competitors to stay competitive,
    - Understanding customers' purchase behavior and finding demanded products,
    - Determining special users such as whales that bring significant income,
    - Executing targeted ads,
    - Conducting customer sentiment analysis.

Applied Technologies in Online Shopping (Cont.)

- *Mobile Apps*
  - Customers can interact with companies' brand continuously through apps, and they can learn about fresh and useful purchasing opportunities.
  - Additionally, mobile apps provide customers with a simple check-out process.

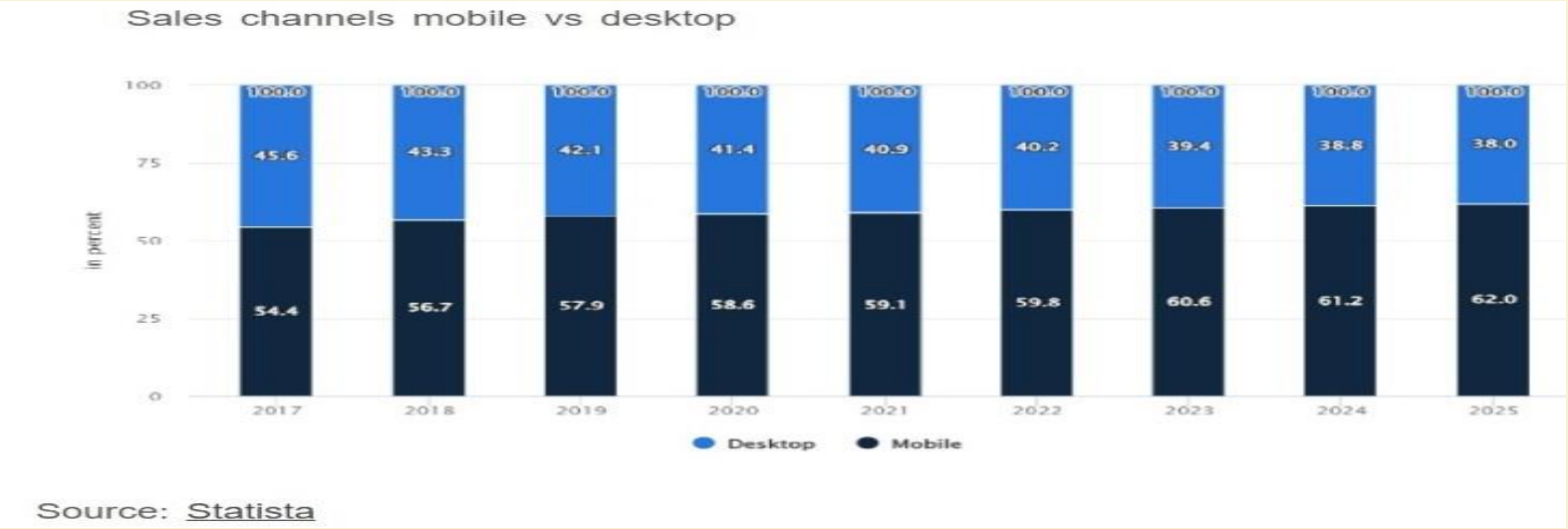


Figure 22: Sales channels mobile and desktop. Source: https://research.aimultiple.com/ecommerce-technology/

Applied Technologies in Online Shopping (Cont.)

- *Supply chain visibility software*
  - The entire supply chain, from delivery to warehouses, may be managed using supply chain visibility tools.
  - These systems automate the issuance of invoices and track inventory.
  - Supply chain visibility software also helps to produce reports that can support the decision-making process.
  - They generate financial reports and some supply chain management systems have the ability to calculate product carbon footprint of companies.

Applied Technologies in Online Shopping (Cont.)

- *Warehouse automation technologies*
  - Warehouse automation provides benefits such as
    - Accurate scheduling of intermediate goods and final products
    - Optimal inventory management
    - Easier financial/environmental reporting
    - Carbon footprint reduction and more.

Applied Technologies in Online Shopping (Cont.)

- *Warehouse automation technologies*

  - Ecommerce companies can automate their warehouses with the following technologies

    - IoT: Optimize the storage conditions of the products while minimizing energy cost.
    - Intelligent automation: Combine Natural Language Processing (NLP) and Robotic Process Automation (RPA) technologies. Order processing is automated by these technologies.
    - Cobots: Work around the clock and package and palletize products.

Applied Technologies in Online Shopping (Cont.)

- *E-wallets*
  - An e-wallet, a kind of electronic card, is connected to a person's bank account.
  - It is utilized for online purchases on a computer or a smartphone.
  - Two major components of an e-wallet are software and data.
  - E-wallets have three main benefits:
    - They increase security.
    - They ease the payment process.
    - Ease of refund which takes a few seconds only.

Applied Technologies in Online Shopping (Cont.)

- *Check-out free systems*

  - Check-out free systems help customers make in-store purchases without queuing with technologies

    - Computer vision.
    - Machine learning algorithms.
    - Smart sensors.
    - E-wallets or custom mobile apps.

Applied Technologies in Online Shopping (Cont.)

- *Image Search Engine*
  - Customers can use Google Images to search for products by capturing images of them and using reverse image search to find photos that are related to the product.
  - Businesses can take advantage of this by uploading a file for each item they sell on their internet store.
  - Using images effectively is a key potential for businesses to succeed in e-commerce since Google Images is one of the most popular internet search engines.

Applied Technologies in Online Shopping (Cont.)

- *Other technologies*
  - Virtual reality (VR) technology solves that issue by creating 3D artificial environments that users can explore and interact with.
  - For VR, customers can virtually try out products, arrange furniture in their homes, and more.
  - Using augmented reality (AR), customers can preview products and be more likely to pick the right product the first time.
  - …

Online Shopping Statistics

- *General*
  - ❖ Global eCommerce sales are expected to reach $6.86 trillion in 2025, up from $6.3 trillion in 2024.
  - ❖ Around 2.8 billion people worldwide will make at least one online purchase this year.
  - ❖ The Asia-Pacific region continues to lead global online sales, with strong growth in emerging markets such as India and Indonesia.
  - ❖ The average online order value increased by 7% year-over-year, reflecting stronger consumer confidence and broader product variety.

Source: Online Shopping Statistics 2026: Data-Driven Insights for Marketers. Link: https://optinmonster.com/online-shopping-statistics/

Online Shopping Statistics (Cont.)

- *E-commerce stores statistics*
  - There are an estimated 26 to 30 million eCommerce stores worldwide, according to 2025 research from Red Stag Fulfillment and other global analyses.
  - Fewer than one million online stores generate meaningful annual revenue, illustrating how competitive and fragmented the eCommerce landscape has become.
  - Shopify powers over 4.8 million active stores, while WooCommerce and Wix eCommerce continue to dominate small-business adoption.

Online Shopping Statistics (Cont.)

- *E-commerce stores statistics*
  - About 22% of all new websites launched in 2024 included an eCommerce component such as a cart or checkout function, reflecting how store functionality has become a standard part of modern website builds.
  - The average online retailer sells through three or more channels, combining their own site, marketplace listings, and social commerce platforms to reach more customers.

Online Shopping Statistics (Cont.)

- *Abandoned Cart*
  - The average global cart abandonment rate is 69.8%, meaning nearly seven in ten shoppers leave before completing their purchase.
  - On mobile devices, the abandonment rate increases to approximately 85%, making mobile checkout the most common drop-off point in the purchase journey.

Online Shopping Statistics (Cont.)

- *Abandoned Cart*
  - The top reasons shoppers abandon carts include:
    - Unexpected extra costs such as shipping or taxes (48%);
    - Forced account creation (24%)
    - Complicated or lengthy checkout processes (17%)
    - Concerns about payment security (18%)
    - Offering guest checkout can reduce abandonment by up to 35%, while displaying shipping costs early in the process can significantly lower drop-offs.

Online Shopping Statistics (Cont.)

- *Technology*
  - ❖ More than 40% of retailers are already using AI tools for personalization, pricing, and logistics.
  - ❖ AI-powered product recommendations can increase average order value by 15–25%,
  - ❖ Nearly 63% of organizations using generative AI report revenue uplift within deployed business units.
  - ❖ Voice-assisted shopping still represents less than 10% of total online sales, though usage continues to grow in electronics and home categories.
  - ❖ Retailers investing in predictive analytics have reported up to a 20% improvement in customer retention.

Online Shopping Statistics (Cont.)

- *Gen Z and Social Commerce Statistics*
  - There are an estimated 60% of Gen Z shoppers discover products through TikTok, Instagram, or YouTube before visiting an online store.
  - About 37% of Gen Z consumers in the U.S. have made at least one direct purchase through a social media platform.
  - Live shopping adoption exceeds 40% in China, where influencer-led retail has become a dominant part of the eCommerce ecosystem.
  - Nearly 70% of Gen Z shoppers trust creator or influencer recommendations more than brand advertisements, emphasizing the importance of authenticity in online marketing.

In conclusion

- What is Online Shopping?
- Applied Technologies in Online Shopping
- Online Shopping Statistics



THANK YOU
FOR YOUR ATTENTION

Q&A